NBA Home-Win Probability Model

Mathematics & Full Constant Provenance

Deployed champion: Elo-only (architecture `conservative`) — technical briefing

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Prepared for recruiter / technical review.

Every number below is classified **Fixed**, **Selected**, **Fitted**, **Derived**, or **Numerical**, with an explicit source or calculation.

Artifacts: `artifacts/selected_spec.json`, `march_architecture_results.csv`, `coefficient_table.csv`.

Purpose of this document. This note states the model equations and accounts for **every numeric constant**: if a number is calculated, the exact optimization is written; if it is not calculated, the external or design source is named. No constant is left as an unexplained "magic number."

1 Champion model (one equation)

The **deployed champion is Elo-only**: an L2-logistic map on the standardized margin-of-victory Elo rating differential,

$$\boxed{p_g = \sigma\big(c + w\,\tilde{x}_g^E\big)}, \qquad \tilde{x}_g^E = \frac{x_g^E - \mu}{s},$$

with **deployed** coefficients (fit on all games through 2026-03-31) $w = 0.9271823510$, $c = 0.2415428879$, $\mu = 0.1408146877$, $s = 0.2690252334$ (raw-unit weight $w/s \approx 3.4464512$), and $\sigma(z) = 1/(1+e^{-z})$, $\operatorname{logit}(p) = \log(p/(1-p))$. This is exactly the calibrated Elo component of §5; April performance state is frozen at 2026-03-31.

Rejected challenger. An Elo + Bradley–Terry/recent-trend convex logistic stacker, $p_g^{\text{blend}} = \sigma(a \operatorname{logit}(p_g^E) + b \operatorname{logit}(p_g^R) + c)$ with $a, b \geq 0,\ a + b \leq 1$, was implemented and **rejected**: under nested rolling-origin validation (§9) it is worse than Elo-only on both log loss and Brier and is worse calibrated. It is retained only in the `challenger` block of `artifacts/selected_spec.json`; see §3.3.

2 How to read every constant

Class	Meaning
Fixed	Hardcoded design choice. Not estimated on this CSV.
Selected	Chosen by comparing candidates on **March log loss**.
Fitted	Numeric solution of an optimization / MLE on data.
Derived	Algebraic consequence of other constants (no new information).
Numerical	Solver / clip / RNG setting; not a modeling claim.

Honesty rule used below. If a number was not estimated or selected on this project's data, this document says so. In particular, the MOV constant 2.2 is **Fixed** (literature), not **Fitted**.

3 Master ledger — every constant

3.1 A. Elo core (Fixed unless noted)

Constant	Value	Class	Provenance / calculation
Initial rating $R_i(0)$	1500	Fixed	Standard Elo origin. Code: `ratings = {team: 1500.0}` in `features.py`. Not fit.
Elo scale S	400	Fixed	Classical Elo logistic scale: a 400-point gap ⇒ 10:1 odds in the raw Elo formula. Denominator in $x^E = (R_h - R_a + H)/400$.
Base of Elo logistic	10	Fixed	Classical Elo uses base-10 logistic, not e. Formula $1/(1 + 10^{-x})$.
MOV mode	`log`	Selected*	Architecture field `elo_mov`. All five candidates used `log`; mode itself was not raced against `none` in selection.
MOV blowout shape	$\ln(\lvert m\rvert + 1)$	Fixed	FiveThirtyEight / Nate Silver NFL Elo log-MOV form.
MOV constant b_{MOV}	2.2	Fixed (external)	Adopted from the published FiveThirtyEight NFL Elo MOV formula (Silver). **Not re-estimated on this CSV.** Full professional rationale, benefits, and how to compute a data-optimal alternative: §5.4.
MOV slope a_{MOV}	0.001	Fixed	Paired with 2.2 in the same FTE formula. Ratio a/b sets favorite/underdog discount. Not fit here.
MOV denom floor	0.25	Fixed	Code guard `max(0.25, denom)` against pathological denominators. Not estimated.

*Selected only in the weak sense that the winning architecture kept `log`; no candidate used `none`.

3.2 B. Architecture hyperparameters for champion `conservative`

These appear in `configs/architecture_candidates.json` and the winner in `artifacts/selected_spec.json`. **Selection rule:** each procedure (Elo-only, rank-only, blend) minimizes its own March sequential log loss (`nba_wp/selection.py`); the champion is Elo-only. April rows loaded during selection: 0. The champion Elo-only model depends only on the Elo fields K, H, MOV, C_E; the remaining fields govern the rejected challenger.

Constant	Value	Class	Provenance / calculation
K (Elo step)	7.5	Selected	Member of `conservative` candidate. Chosen because it had lowest Elo-only March LL among 5 candidates (Table 7). Not a continuous MLE.
H (home Elo pts)	55	Selected	Same bundle. Equal-team home edge ⇒ $1/(1 + 10^{-55/400}) \approx 0.578$.
Elo logistic C_E	10	Selected	Regularization for the deployed Elo calibrator. From `conservative`.
C_{BT}	0.1	Selected†	Bradley–Terry regularization (challenger only).
Trend half-life $\tau_{1/2}$	60 days	Selected†	EWMA weight $2^{-\Delta\text{days}/60}$ (challenger only).
Trend short window n_S	12 games	Selected†	Mean of last $\min(12, n_i)$ margins (challenger only).
Rank logistic C_R	0.1	Selected†	BT+trend calibrator regularization (challenger only).
Architecture name	`conservative`	Selected	Argmin of Elo-only March LL over 5 named bundles.

†These fields do not enter the deployed Elo-only champion; they parameterize the rejected challenger blend.

3.3 C. Deployed Elo-only coefficients (Fitted)

The deployed champion has **no stacker**: it is the calibrated Elo logistic (§5), fit on all rows through 2026-03-31. From `artifacts/coefficient_table.csv`:

Component	Feature	Std. coef	Raw-unit coef
Elo	x^E = elo_diff	0.92718235	3.44645125
Elo	intercept	0.24154289	—

Raw-unit coef = standardized coef / training scale ($s = 0.26902523$, $\mu = 0.14081469$). Fitted by penalized MLE (`lbfgs`, $C_E = 10$).

Rejected challenger stacker. For completeness, the challenger blend's convex stacker (architecture `conservative`) has **deploy** coefficients $a = 0.3976558706$, $b = 0.6023441294$, $c = 0.3347604843$ ($a + b = 1$, temperature $\tau = 1$), from an unconstrained MLE fit $a = 0.6076317564$, $b = 0.9204024091$, $c = 0.3279170537$ ($\tau = 1/(a + b) \approx 0.655 < 1$). `fit_logit_stacker` enforces a genuine convex blend — it clips any negative weight to zero and applies the temperature floor $\tau \geq 1$, so $0 \leq w \leq 1$ for both components (pinned by `tests/test_stacker_temperature_floor.py::test_stacker_weights_are_convex_when_stacker_is_used`). This blend is not deployed; it loses to Elo-only out-of-sample (§9). The full challenger coefficients live in the `challenger` block of `artifacts/selected_spec.json`.

3.4 D. Challenger base-component coefficients (Fitted, reference)

The rejected challenger's rank component (Bradley–Terry + trend) is an L2-logistic calibrator; its coefficients are archived with the challenger. They are not part of the deployed champion, which uses only the Elo row in §C.

How calculated. For training rows available at that stage,

$$\hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \sum_i \left[Y_i \log \sigma(\beta_0 + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i) + (1 - Y_i) \log(1 - \cdots) \right] - \tfrac{1}{2C} \|\boldsymbol{\beta}\|_2^2,$$

with $\tilde{\mathbf{x}}$ standardized (`StandardScaler`), $C = C_E$ or C_R, solver `lbfgs`. Raw-unit coef = standardized coef / training scale.

3.5 E. Bradley–Terry / trend auxiliaries

Constant	Value	Class	Provenance
BT design coding	$+1/-1$	Fixed	Home column $+1$, away -1. Standard BT contrast.
BT team strengths q_i, α	(vector)	Fitted	L2-logistic MLE on all prior games before date d, $C = C_{\mathrm{BT}}$. Refit when history grows. Feature $x^{\mathrm{BT}} = \hat{\alpha} + \hat{q}_h - \hat{q}_a$.
EWMA base 1/2	0.5	Fixed	Half-life definition: weight $2^{-\Delta/\tau_{1/2}} = (1/2)^{\Delta/\tau_{1/2}}$.
Beta prior (α, β)	$(4, 4)$	Fixed	Comment in code: "eight games of neutral information." Smoothed win rate $(W + 4)/(G + 8)$. Used for `record_logit` (ablation feature; not in champion X).
Rest cap	7 days	Fixed	Rest days clipped to $[0, 7]$. Schedule features only (ablation path).
B2B threshold	1 day	Fixed	`back_to_back = (rest <= 1)`. Ablation path.
Density windows	4, 6 days	Fixed	Games-in-window counts. Ablation path.

3.6 F. Calendar / selection policy (Fixed policy)

Constant	Value	Class	Provenance
Train-through (base, March scoring)	2026-02-28	Fixed policy	`selection_policy.json` / selection code.
Selection window	2026-03-01–03-31	Fixed policy	Same.

Constant	Value	Class	Provenance
April holdout start	2026-04-01	Fixed policy	Contract: selection raises if April present.
Accuracy cutoff	0.5	Fixed	`p >= 0.5` in metrics. Not optimized.
Primary metric	log loss	Fixed policy	Selection rule text.
# architectures raced	5	Fixed design	Length of candidate list.

3.7 G. Derived quantities (Derived)

Quantity	Value	Calculation
Elo raw-unit weight	3.4465	$w/s = 0.92718235/0.26902523$ (deployed champion)
Challenger $w = a/(a+b)$ (raw)	0.3977	$0.60763176/(0.60763176 + 0.92040241)$; preserved by the floor
Challenger τ (raw)	0.6551	$1/(0.60763176 + 0.92040241) < 1 \Rightarrow$ floor applied
Challenger τ (deploy)	1.0000	$1/(0.39765587 + 0.60234413)$ after projection onto $a + b = 1$
Equal-team home p	≈ 0.578	$1/(1 + 10^{-H/400})$ with $H = 55$
Fair home odds	$1/p_g$	Definition
Fair away odds	$1/(1 - p_g)$	Definition

3.8 H. Numerical / solver settings (Numerical)

Setting	Value	Where
`max_iter`	10000	all logistics
`tol`	10^{-12}	all logistics
`random_state`	0	reproducibility
`solver`	`lbfgs`	all logistics
probability clip (logit)	$[10^{-12}, 1 - 10^{-12}]$ or $[10^{-9}, 1 - 10^{-9}]$	`model.py/features.py`

4 Target and information

$$Y_g = \begin{cases} 1, & \text{home points}_g > \text{away points}_g, \\ 0, & \text{otherwise.} \end{cases} \qquad X_g = f(\mathcal{H}_{<d}).$$

Same-date batching: write all rows for date d before any Elo/BT/trend update from results on d.

5 Elo mathematics (the deployed champion)

5.1 Raw probability and feature

$$p_g^{\text{Elo,raw}} = \frac{1}{1 + 10^{-(R_h - R_a + H)/400}}, \qquad x_g^E = \frac{R_h - R_a + H}{400}.$$

Constants: $R_i(0) = 1500$ (Fixed), 400 (Fixed), $H = 55$ (Selected).

5.2 Update with MOV

$$R'_h = R_h + KM_g(Y_g - p_g^{\text{Elo,raw}}), \qquad R'_a = R_a - KM_g(Y_g - p_g^{\text{Elo,raw}}),$$
$$M_g = \ln(|m_g| + 1) \cdot \frac{2.2}{\max\big(0.25,\, 0.001(R_{\text{win}} - R_{\text{lose}}) + 2.2\big)},$$

where $R_{\text{win}}, R_{\text{lose}}$ are the pregame ratings of the *winning* and *losing* team (§5.3). Constants: $K = 7.5$ (Selected); 2.2, 0.001, 0.25 (Fixed).

5.3 MOV rating-difference sign (winner − loser)

The rating difference inside the MOV denominator is the *winner minus loser* rating, not a fixed *home minus away* rating. This matches the published FTE form ($\text{ELO}_W - \text{ELO}_L$) and has two correct consequences: (i) an **upset** (a lower-rated team winning) gives a smaller denominator and hence a *larger* rating update than an expected result, and (ii) with zero home advantage the update is exactly team-swap symmetric. Using a fixed home−away difference instead would make the autocorrelation correction depend on venue rather than on who actually won, under-updating home upsets and over-updating home favorites. This behaviour is pinned by `tests/test_elo_mov_winner_diff.py`.

5.4 The constant 2.2: source, case for use, and how to optimize it

Disclosure (external source). The value 2.2 was **not estimated from this project's NBA dataset**. It was adopted from a published, widely cited sports-rating specification: Nate Silver / FiveThirtyEight, *Introducing NFL Elo Ratings* (archived: https://web.archive.org/web/20171004105151/https://fivethirtyeight.com/features/introducing-nfl-elo-ratings/), which defines

$$M = \ln(|PD| + 1) \cdot \frac{2.2}{(\text{ELO}_W - \text{ELO}_L) \cdot 0.001 + 2.2}.$$

This submission copies that functional form into `nba_wp/features.py::_elo_multiplier` when `elo_mov="log"`. Transparency on this point is intentional: industry models routinely reuse validated structural constants when the *form* is what matters and the local sample is too short to re-estimate every shape parameter cleanly.

Professional case for using 2.2 here (best case, optimal or not). Even though 2.2 is external, using this specific published constant is a defensible engineering choice for a leakage-audited pricing model:

1. **Correct economic shape, not an arbitrary fudge.** The log term gives diminishing returns to blowouts (a 30-point win should not update ratings like three independent 10-point wins). The fraction involving 2.2 implements the standard underdog/favorite correction so favorites that "run up the score" do not inflate permanently. That is precisely the behavior a book wants in a strength tracker feeding probabilities.
2. **Stability and reproducibility.** A public constant with a fixed formula is auditable: two reviewers regenerating features get the same Elo path. Replacing it with an in-sample tuned value on one partial season would add a free parameter, widen the selection surface, and invite overfit to March noise.
3. **Separation of concerns in the validation design.** This project deliberately concentrates selection effort on quantities that move prices most (home advantage H, step size K, BT/trend regularization) under a March log-loss rule, while holding the MOV *shape* at a known industry default. That is a standard bias–variance trade: fix low-sensitivity shape parameters; select high-sensitivity ones.
4. **Empirical support that the choice is not fragile.** Varying the constant in a neighborhood of 2.2 changes the Elo-only March log loss only in the fourth decimal place (flat likelihood). So the *presence* of this MOV correction is beneficial; the exact published 2.2 is a stable operating point, not a knife-edge claim of global optimality.
5. **Interpretability for stakeholders.** 2.2 has a clear reading in the literature as the baseline expected $\ln(|PD|+1)$ scale in the autocorrelation correction (see Morse's derivation of $d = 2200 = 2.2/0.001$). That is preferable to an opaque tuned scalar with no external meaning.

How to calculate the value that would have been most optimal. If the goal is a *data-optimal* MOV constant $b^\star$ for this pipeline (rather than the published 2.2), compute it by the same selection principle already used for architectures—minimize March sequential log loss—while holding the champion architecture fixed:

1. Fix architecture `conservative` except replace the literal 2.2 by a candidate $b > 0$ in

$$M_g(b) = \ln(|m_g| + 1) \cdot \frac{b}{\max\bigl(0.25,\, 0.001(R_h - R_a) + b\bigr)}.$$

   (Optionally keep the literature ratio by setting the slope to $b/2200$ instead of 0.001; the one-parameter profile below is the minimal change.)
2. For each candidate b on a grid (e.g. $b \in \{1.0, 1.5, \ldots, 4.0\}$) or by continuous search: build features → fit base models on games before 2026-03-01 → fit the March stacker → record March log loss $\mathrm{LL}(b)$.
3. Set the optimal replacement

$$\boxed{b^\star \;=\; \arg\min_b \; \mathrm{LL}_{\mathrm{March}}(b)}$$

   That is: choose the MOV constant b that minimizes March sequential log loss. This $b^\star$ is the March-optimal MOV constant for this model and metric. On a quick profile of the current code path, $\mathrm{LL}(b)$ was minimized toward the high end of the tested grid (near $b = 4$), with only a tiny improvement over 2.2 ($\Delta\mathrm{LL} \sim 4 \times 10^{-5}$). Hence 2.2 is not claimed as $b^\star$; it is claimed as a published, stable default whose March performance is essentially tied with nearby optima.
4. **Stronger production estimator (multi-season).** On a longer NBA history, estimate the literature intercept directly: regress bin-wise $\widehat{\mathbb{E}}[\ln(|PD| + 1) \mid \text{win}, x]$ on Elo gap x and take the intercept $\hat{b}_0$ (and slope $\hat{a}_0$) as in the Morse/FTE construction, then set $b \leftarrow \hat{b}_0$. Alternatively, nest b inside the architecture search and select by prequential log loss with a locked final holdout.

One-sentence position for reviewers. *I used the published FiveThirtyEight log-MOV constant* 2.2 *as a fixed, auditable shape parameter because it encodes a validated blowout/underdog correction, keeps the Elo path reproducible, and sits on a flat region of the March likelihood; a March (or multi-season) profile of* $\mathrm{LL}(b)$ *is the correct way to replace it with a data-optimal* $b^\star$ *if that becomes a production requirement.*

5.5 Calibrated Elo probability (the deployed price)

$$p_g^E = \sigma(\beta_0^E + \beta_1^E \tilde{x}_g^E) = p_g, \quad C_E = 10 \text{ (Selected)},$$

with deployed $\beta_0^E = 0.2415428879$, $\beta_1^E = 0.9271823510$ (standardized). This *is* the champion of §1.

6 Bradley–Terry + trend (rejected challenger only)

The following components feed only the rejected challenger blend, not the deployed Elo-only champion.

6.1 Bradley–Terry

Prior games before date d:

$$P(Y = 1) = \sigma(\alpha + q_h - q_a), \quad C_{\mathrm{BT}} = 0.1 \text{ (Selected)}.$$

Feature: $x_g^{\mathrm{BT}} = \hat{\alpha} + \hat{q}_h - \hat{q}_a$ (**Fitted** each day).

6.2 Trend

$$L_{i,d} = \frac{\sum_{k<d} 2^{-\Delta_{kd}/60} m_{ik}}{\sum_{k<d} 2^{-\Delta_{kd}/60}}, \qquad S_{i,d} = \frac{1}{n_i^{\mathrm{short}}} \sum_{\text{last } n_i^{\mathrm{short}}} m_{ik},$$

$$n_i^{\mathrm{short}} = \min(12, n_i), \quad T_{i,d} = S_{i,d} - L_{i,d}, \quad x_g^{\mathrm{trend}} = T_{h,d} - T_{a,d}.$$

Constants 60 and 12: **Selected** with `conservative`. Base 2 in $2^{-\cdot}$: **Fixed** half-life algebra.

6.3 Rank component

$$p_g^R = \sigma(\beta_0^R + \beta_{\mathrm{BT}}\tilde{x}_g^{\mathrm{BT}} + \beta_{\mathrm{tr}}\tilde{x}_g^{\mathrm{trend}}), \quad C_R = 0.1 \text{ (Selected)}.$$

7 Selection proof for architecture constants

Elo-only March sequential log loss (base fit through February) from `artifacts/march_architecture_results.csv` (lower LL wins). The champion is chosen by this Elo-only surface:

Architecture	K	H	Elo-only March LL	Winner
conservative	7.5	55	**0.50659003**	yes
hfa_75	10	75	0.50909006	
balanced	10	65	0.50913895	
lower_bt_shrinkage	10	65	0.50913895	
responsive	15	65	0.51470876	

This is the **entire calculation** behind calling $K = 7.5$, $H = 55$, $C_E = 10$ "selected" for the deployed Elo-only champion: they are the coordinate values of the bundle with minimal Elo-only March LL. They are not independently profiled. (Each other procedure selects its own architecture by its own March LL; see `march_architecture_results.csv`.)

8 Metrics (definitions; cutoffs Fixed)

$$\mathrm{LL} = -\frac{1}{N}\sum_g \left[Y_g \log p_g + (1 - Y_g)\log(1 - p_g)\right], \quad \mathrm{BS} = \frac{1}{N}\sum_g (p_g - Y_g)^2.$$

Accuracy uses Fixed cutoff $p_g \geq 0.5$.

Champion results (Elo-only, artifacts). **Primary April holdout (frozen pre-April)**: LL 0.464369, Brier 0.149770, AUC 0.866847, accuracy 78.13% ($N = 96$). March selection period (Elo-only, base fit through February, $N = 239$): LL 0.506590. Optional April sequential backtest (live-update simulation): LL 0.464648. Rejected blend on the same frozen April window: LL 0.468725, Brier 0.150465 — worse on both.

9 Nested rolling-origin validation (why Elo-only)

`scripts/nested_validation.py` validates the whole procedure under two information policies (frozen-block and daily-sequential; never mixed), 11 weekly outer folds, 501 out-of-sample games. Each procedure independently selects its own architecture by its own inner out-of-fold score; the challenger stacker trains on inner out-of-fold predictions. Pooled out-of-sample (frozen-block): Elo-only LL 0.532/Brier 0.177; rank-only 0.550/0.184; blend 0.548/0.183; constant 0.688/0.247. Block-bootstrap blend−Elo: $\Delta\mathrm{LL} = +0.017$ (95% CI $[+0.010, +0.023]$), $\Delta\mathrm{Brier} = +0.006$ (95% CI $[+0.004, +0.009]$); **0 of 4,000** week-block replicates favored the blend. Calibration (frozen-block): Elo-only slope $\beta \approx 1.37$ (95% CI $[1.22, 1.57]$), ECE ≈ 0.059 — better calibrated than the blend ($\beta \approx 1.80$). Decision under both policies: **keep Elo-only**. Artifacts: `nested_frozen_block_summary.json`, `nested_daily_sequential_summary.json`.

10 End-to-end constant flow

Deployed champion (Elo-only):

$$\underbrace{1500, 400, 2.2, 0.001}_{\text{Fixed Elo}} + \underbrace{K, H, C_E}_{\text{Selected}} \xrightarrow{\text{features}} x^E \xrightarrow[\text{Fitted}]{c,\, w} p^E = p.$$

Code map. `nba_wp/features.py` (Fixed Elo/MOV; Selected architecture fields), `nba_wp/model.py` (Fitted Elo c, w; challenger a, b, c), `nba_wp/selection.py` (model-specific selection via March LL), `artifacts/selected_spec.json` (persisted Elo-only champion + rejected challenger).